

Introductory problems worked solutions

MaTCo Session 1

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Group Problem 3.

Problem. Slips containing the names of n people are shuffled and handed out to them at random. What is the probability that any person received their own name?

Solution. Let us denote the set of people in the room by the set $N = \{1, 2, \dots, n\}$, where each number from 1 to n is used to label or represent a different person/name.

Let us denote by the symmetric group S_n the set of permutations $\sigma : N \rightarrow N$ that could have been used by the shuffling process to assign each person a name. Concretely, if person a received the name of person b , then we say $\sigma(a) = b$.

We know that there are $n!$ possible permutations in S_n , since for $\sigma(1)$ there are n choices of possible names, for $\sigma(2)$ there are $n - 1$ choices, and so on. From these $\prod_{i=1}^n i = n!$ permutations, one is selected uniformly. To find the probability that no one received their own name, we have to find the proportion of permutations σ such that for each person $i = 1, 2, \dots, n$, $\sigma(i) \neq i$, i.e. that σ does not have any [fixed points](#).

Such a permutation is called a [derangement](#), and we denote the number of derangements in S_n by $!n$. The proportion of permutations that are derangements is $\frac{!n}{n!}$, and we will find both a more concrete expression for this, as well as an approximation that is fascinatingly not a function of n . This tells us something about the structure of derangements: they make up a nearly constant proportion of all permutations.

For any derangement $\sigma \in S_n$, let $\sigma(n) = k$, where $k \in 1, \dots, n - 1$. This approach lets us break all the derangements into two cases:

Case 1: $\sigma(k) = n$, so n and k map to each other, or form a [cycle](#). Then if we remove this cycle by removing the elements k and n from the domain and range of σ , we are left with a derangement of the set $N \setminus \{k, n\} = \{1, \dots, k - 1, k +$

$1, \dots, n-1\}$, of which there are $!(n-2)$ possible such derangements.

Case 2: $\sigma(k) \neq n$. Let j be the unique person such that $\sigma(j) = n$, with $j \neq k$. Then if we redefine $\sigma(j) := k$ and remove n from the domain of σ , then we are left with a derangement of the set $\{1, \dots, n-1\}$, of which there are $!(n-1)$ such derangements.

We can partition the set of all derangements of n people based on the choice of k : For each k we obtain a subset of $!(n-2) + !(n-1)$ derangements from the cases above, and there are $n-1$ possible values that k can take, giving us the recursion $!n = (n-1)(!(n-2) + !(n-1))$, o.e. $!(n+1) = n \cdot !n + n \cdot !(n-1)$ for all $n \geq 1$.

To solve this recursion we introduce a technique: let $F(x) = \sum_{n=1}^{\infty} (!n) \frac{x^n}{n!} = (!1) \cdot x + (!2) \cdot \frac{x^2}{2} + (!3) \cdot \frac{x^3}{3!} + \dots$ denote the [exponential generating functions](#) of the sequence $\{!n\}_{n=0}^{\infty}$ of the number of derangements. The introduction and usage of this abstract function may appear unusual, but we will go on to see why this technique can be very useful for solving problems pertaining to recursion or combinatorics. You may have encountered something similar in the form of Taylor series, power series, or the moments of a random variable!

Notice that we can represent sequences for terms in the recursion with generating functions by modifying $F(x)$: differentiating the series with respect to x moves the power of every term down and hence gives it the coefficient that previously belonged to the term above it (with power 1 greater). Multiplying the series by x shifts the exponent of every term up, leaving it with a coefficient one below its usual, but due to the factorial denominator, also multiplies each term by n .

This means that the generating functions for the sequences $\{!(n+1)\}_{n=1}^{\infty}$, $\{n \cdot !(n-1)\}_{n=2}^{\infty}$ and $\{n \cdot !n\}_{n=1}^{\infty}$, are $F'(x)$, $xF(x)$, and $xF'(x)$ respectively.

These 3 generating functions let us redescribe our recursion over all n and draw some conclusions from it.

$$\begin{aligned} F'(x) &= xF'(x) + xF(x) \\ (1-x)F'(x) &= xF(x) \\ F'(x) &= \frac{x}{1-x} F(x) \\ \frac{F'(x)}{F(x)} &= \frac{x}{1-x} \end{aligned}$$

We now have a differential equation for $F(x)$, which we can solve by noticing that $\frac{F'(x)}{F(x)} = (\log |F(x)|)'$, where $\log$ denotes the natural logarithm.

$$\begin{aligned}
\log |F(x)| &= \int \frac{x}{1-x} dx \\
&= \int \left(\frac{1}{1-x} - 1 \right) dx \\
&= -\log |1-x| - x + c \\
F(x) &= A \cdot (e^{-1})^{\log |1-x|} \cdot e^{-x} \\
&= \frac{Ae^{-x}}{1-x}
\end{aligned}$$

Here A is a constant, which we can find by imposing a boundary condition that $F(0) = !0$. There is exactly one permutation of the empty set, which is from the empty set to the empty set, and this permutation cannot have any fixed points, so it must be a derangement and hence $!0 = 1$.

This gives us $\frac{Ae^{-0}}{1-0} = A = 1$, so $F(x) = \frac{e^{-x}}{1-x}$. From the Taylor Series expansions $e^{-x} = \sum_{n=0}^{\infty} (-1)^n x^n / n!$ and $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, we have that the Cauchy product of these power series is given by:

$$\begin{aligned}
\sum_{n=0}^{\infty} (!n) \frac{x^n}{n!} &= \left(\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!} \right) * \left(\sum_{n=0}^{\infty} x^n \right) \\
&= \sum_{n=0}^{\infty} \left(\sum_{k=0}^n \frac{(-1)^k}{k!} \right) x^n \\
\frac{!n}{n!} &= \sum_{k=0}^n \frac{(-1)^k}{k!} \\
&\approx e^{-1}
\end{aligned}$$

We can more precisely show that $\sum_{k=0}^n (-1)^k / k!$ rapidly approaches $1/e$ as n approaches ∞ :

It holds that $|1/e - \sum_{k=0}^n (-1)^k / k!| \leq \sum_{k=n+1}^{\infty} |(-1)^k| / k! \leq 1/(n+1)!$, and this sequence is approaching 0 at a rate faster than exponential, so $!n/n!$ comes extremely close to $1/e$ even for relatively small n .

$$\lim_{n \rightarrow \infty} \frac{1}{(n+1)!} = 0 \implies \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k}{k!} = \frac{1}{e}$$

□

This problem can also be solved more easily using inclusion-exclusion, however if you understood the above proof you have now been introduced to a powerful new technique!

Group Problem 4.

Problem. You are playing a game with a friend involving 2 piles of stones. On each turn, a player can remove either an arbitrary number of stones from one pile, or the same number of stones from both piles. Whoever removes the last stone is the winner. Find all pairs (x, y) of the initial number of stones on each pile, such that the second player is winning.

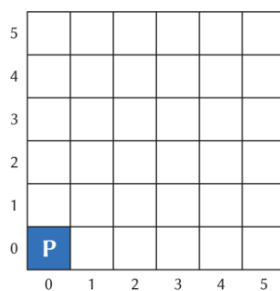
Solution. We call a position in a game winning for a player if, with perfect play, that player is guaranteed to win from that position. In a game like this where one player can win if and only if the other loses, all players know the possible moves the others have available to them and the outcomes of these moves, and there is no randomness, every position must be winning for one player and losing for the others, or a draw for all. Such a game is called a deterministic, perfect information, zero-sum game. In this case the game is 2-player, and a draw is not possible, so one player must win and the other must lose.

Each position can be represented easily by a pair of non-negative integers (x, y) representing the number of stones on each pile, as in each move the only thing that changes is this pair and the player whose turn it is (something we will also need to keep track of), and the game ends if and only if we reach a certain position (x, y) , namely $(0, 0)$.

We can start the game from any such position and have to determine which of them are winning for the second player. We will call such a position losing and denote it by P , as it is losing for the first player, so if a (x, y) is losing we say $(x, y) = P$, and otherwise $(x, y) = N$. We will hence have to consider and label all positions from $\mathbb{Z}_{\geq 0}^2$.

Notably, stones must be taken from at least one of the piles in each move, and they can never be added back. This means that in each turn, the value of $\max(x, y)$ must decrease by an integer amount, so the position (x, y) will inevitably be reached. This proves that a draw is not possible.

To try and get a feel for the problem, we can start with some simple cases, and to encourage geometric thinking we will draw the space of positions as a chessboard stretching out to infinity.

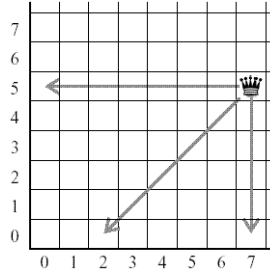


If the position is $(0,0)$ at the start of a player's turn, that player has lost, so if the starting state is $(0,0)$ we should say it is losing, or $(0,0) = P$. From the positions $(1,0)$ or $(0,1)$ the first player can win by taking the remaining stone from one of the piles, so those positions are winning. From $(1,1)$ the first player is also winning as they can take one stone from both piles. In fact, any position of the kind $(0,n)$, $(n,0)$, or (n,n) is winning, where $n \in \mathbb{Z}^+$, as the first player can take n stones from one or both piles to win.

Let's consider the position $(2,1)$: The first player has to move to one of $(2,0)$, $(1,1)$, $(0,1)$, or $(1,0)$. We established that all of these moves are winning for the player who starts at that position, so if the first player starts at $(2,1)$, they are forced to put the second player in a winning position, and are as such losing. By symmetry along the diagonal, this is also true for $(1,2)$. Since

We can swap our piles of stones around w.l.o.g. (without loss of generality), so we say that $(x,y) = (y,x)$. Hence by symmetry across the diagonal, $(1,2)$ is also losing, or $(1,2) = (2,1) = P$.

We noticed that moves on the chessboard are like those of a queen: horizontally, vertically or diagonally, although only to the left, downward or both. A position is losing if and only if all such moves from that position lead to a winning position.



Any position that is directly above, directly to the right of, or diagonally above and to the right of a losing position (x,y) must be winning, since from any of those positions the first player can move to (x,y) , which is now losing for the second player and forcing a win for the first. Hence the queen visualization is apt, as all moves along the same row, column or leading diagonal of a losing position are winning. Hence we say that $(x,y) = P$ if and only if for all other positions (a,b) such that $a = x$, $b = y$ or $y - x = b - a$, $(a,b) = N$.

We can iteratively construct a sequence $\{(x_n, y_n)\}_{n=0}^{\infty}$ of losing positions, starting with $(0,0)$ and $(1,2)$. We will assume w.l.o.g. that $x_n < y_n$, since $(x_n, y_n) = (y_n, x_n)$, and x_n cannot equal y_n else the position would lie on the same diagonal as $(0,0)$ and be winning. We are also assuming that there are infinitely many losing positions, which we will see from our sequence construction is the case.

No position in our sequence can share a coordinate x_n or y_n with any of the

positions that came before it, since that would mean lying on the same row or column as one of the previous losing positions (x_i, y_i) in the sequence, or on the same row or column as one of the losing reflections (y_i, x_i) .

Indeed, x_n will take the smallest value not yet used by any other coordinates, since by picking y_n such that (x_n, y_n) does not lie on the same diagonal as a previous losing position, all positions along the same row, column and diagonal as (x_n, y_n) will be winning. As such, $x_n = \min \left(\mathbb{Z}_{\geq 0} \setminus \bigcup_{i=0}^{n-1} \{x_i, y_i\} \right)$, which we also denote as $x_n = \text{mex}(x_0, y_0, \dots, x_{n-1}, y_{n-1})$.

It follows that y_n must take the minimal value greater than x_n such that no previous positions lie on the same diagonal as (x_n, y_n) , o.e. $y_n - x_n = \text{mex}(y_0 - x_0, \dots, y_{n-1} - x_{n-1})$.

These values y_n can be found inductively to be $x_n + n$. When $n = 0$, we have $y_0 = 0 = x_0 + 0$. Assuming that $y_i = x_i + i$ for all $m < n$, it holds that $y_n = x_n + \text{mex}(y_0 - x_0, \dots, y_{n-1} - x_{n-1}) = x_n + \text{mex}(0, \dots, n-1) = x_n + n$.

Looking again at the construction of the sequence x_n , which we want to find a closed form for, we observe that every positive integer takes exactly one value from either x_n or y_n . Let $\sigma : \mathbb{Z}^+ \rightarrow \{0, 1\}$ be an indicator function for this: if a positive integer k is in the sequence x_n , we say $\sigma(k) = 0$. Otherwise, k must be in y_n and $\sigma(k) = 1$. We want to understand the binary word given by $\sigma(1), \sigma(2), \sigma(3), \sigma(4), \sigma(5), \dots = 0, 1, 0, 0, 0, \dots$, which we will denote more simply by $\sigma(1, \dots, 5) = 01001$.

We can also show by contradiction that successive terms in x_n cannot be more than two integers apart. For x_n and x_{n+1} to have a difference greater than 2, $\sigma(x_n + 1) = \sigma(x_n + 2) = 1$. This means there must exist $i, j < n$ such that $x_n + 1 = y_i = x_i + i$ and $x_n + 2 = y_j = x_j + j$. Combining these equations, we get $x_j - x_i + j - i = 1$. This leads to a contradiction, since x_n is strictly increasing. Either: $i < j$, in which case $x_j - x_i \geq 1$ and $j - i \geq 1$ which gives us $x_j - x_i + j - i \geq 2$. Otherwise, $j < i$, in which case $x_j - x_i + j - i \leq -2$, or $i = j$, in which case $x_j - x_i + j - i = 0$.

Let $d_n = x_{n+1} - x_n$ denote the sequence of differences between successive terms in x_n , each of which is equal to either 2 or 1. When we have m such that $d_m = 2$, $x_m + 1$ cannot be in x_n , so $\sigma(x_m + 1) = 1$.

One thing that pops out is that, when computing values of the sequence $\{d_n\}_{n=1}^{\infty} = 2, 1, 2, 2, 1, \dots$, it is exactly the same as the sequence given by $\{2 - \sigma(n)\}_{n=1}^{\infty}$, i.e. each 2 in d_n corresponds to a 0 in the word given by σ , and each 1 corresponds to a 1.

This is demonstrably the case: when $d_m = 2$, $\sigma(x_m + 1) = 1$, so $x_m + 1 = y_k$ for some $k \leq m \in \mathbb{Z}^+$. In the interval $1, \dots, x_m$, there are $k - 1$ terms from y_n , and the remainder of the terms belong to x_n , so $m = x_m - (k - 1) = y_k - k = x_k$ gives the index m of the last term from x in this interval. This gives us a recurrence relation $x_{x_k} + 1 = y_k$, but also tells us that if $d_m = 2$, $m = x_k$ for some $k \leq m$, so $\sigma(m) = 0$.

Now we show the converse that $d_n = 2$ only if $\sigma(n) = 0$. Taking arbitrary $m \in \mathbb{Z}^+$, we assume that $\sigma(m) = 0$, so there exists $k \leq m$ such that $x_k = m$. The earlier recurrence relation tells us that $x_m + 1 = x_{x_k} + 1 = y_k$, so there is a gap between x_m and the next term x_{m+1} which must equal $x_m + 2$. Having shown that $d_n = 2 \iff \sigma(n) = 0$, the only other possible value of d_n and $\sigma(n)$ are 1 so $d_n = 1 \iff \sigma(n) = 1$, and the two binary words are isomorphic, or essentially the same.

The interesting part about this isomorphism is that the terms in d_n also determine future values of $\sigma(m)$ for some $m \geq n$ by indicating gaps between 0's, so the word $\sigma(1, \dots)$ recursively defines itself. Indeed, starting with the word $\sigma(1) = 0$, we know that $d_1 = 2$ so 1 can be concatenated to the end of the word, yielding, $\sigma(1, 2) = 01$ as the successive word. We will denote words and their lengths by W_n and L_n respectively, and the successive word that is generated is W_{n+1} with length L_{n+1} .

Algorithm 1 Generate the successive word W_{n+1}

Require: $W_n = w_1 w_2 \dots w_{L_n} = \sigma(1, \dots, L_n)$

Ensure: $W_{n+1} = \sigma(1, \dots, L_{n+1})$

for $i \leftarrow 1$ to L_n **do**

Invariant: $W_{n+1} = \sigma(1, \dots, x_i - 1)$ at the beginning of iteration i .

$\triangleright \sigma(x_i)$ needs to be concatenated as the next term to W_{n+1}

if $w_i = 0$ **then**

$\triangleright d_i = 2$

$\triangleright \sigma(x_i + 1) = 1$ also needs to be concatenated

$W_{n+1} \leftarrow W_{n+1} 01$

else

$\triangleright d_i = 1$

$\triangleright$ There is no gap proceeding x_i , so no other term

$W_{n+1} \leftarrow W_{n+1} 0$

end if

end for

This algorithm of iteratively substituting $0 \rightarrow 01$ and $1 \rightarrow 0$, beginning with $W_1 = 0$, may seem familiar: it is a definition for the celebrated Fibonacci word, and we aim to find a closed form for (x_i, y_i) by showing some properties of this word.

Firstly, the loop invariant $W_{n+1} = \sigma(1, \dots, x_i - 1)$ tells us that at the end of the algorithm, $W_{n+1} = \sigma(1, \dots, x_{L_n+1} - 1)$, and the algorithm ensures that $W_{n+1} = \sigma(1, \dots, L_{n+1})$, so $x_{L_n+1} - 1 = L_{n+1}$, implying that there are L_n terms from the sequence x_i in W_{n+1} , excluding x_{L_n} but including 0.

When generating the subsequent word W_{n+2} , each of the terms in W_{n+1} will contribute a 0, whilst the L_n terms in $x_0, \dots, x_{L_n-1}$ will also contribute an additional term 1. This means that $L_{n+2} = L_{n+1} + L_n$, so the sequence $\{L_n\}_{n=1}^\infty$ is precisely the Fibonacci sequence. The stronger result $W_{n+2} = W_{n+1} W_n$ is the definition of the Fibonacci word that gives it its name.

Of the L_{n+1} terms in W_{n+1} , L_n are from $x_0, \dots, x_{L_n-1}$, so the remaining $L_{n+1} -$

$L_n = L_{n-1}$ terms must be from $y_0, \dots, y_{L_{n-1}-1}$. For a large word $\lim_{n \rightarrow \infty} W_n$, the proportion of positions of terms in x_n is given by:

$$\lim_{n \rightarrow \infty} \frac{L_{n-1}}{L_n} = \frac{1}{\varphi}$$

Here $\varphi = \frac{1+\sqrt{5}}{2}$ is the golden ratio. This is a famous result of the Fibonacci sequence, however, even without this knowledge we could compute φ directly from our setup.

We know that the proportion of terms $x_0, \dots, x_{L_n}$ in $W_{L_{n+1}}$ converges fairly quickly to some value $1/\varphi$, and that x_n must grow roughly linearly, as the difference between successive terms must be either 1 or 2 and thus $n < x_n < 2n$. Hence it is reasonable to say that $x_n \approx n\varphi$.

We can find the proportion of terms $y_0, \dots, y_{L_n}$ in $W_{L_{n+2}}$ to be

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{L_n}{L_{n+2}} &= \lim_{n \rightarrow \infty} \left(\frac{L_n}{L_{n+1}} \cdot \frac{L_{n+1}}{L_{n+2}} \right) \\ &= \lim_{n \rightarrow \infty} \frac{L_n}{L_{n+1}} \cdot \lim_{n \rightarrow \infty} \frac{L_{n+1}}{L_{n+2}} \\ &= \frac{1}{\varphi^2} \end{aligned}$$

$y_n = x_n + n$, so if we are assuming that x_n grows roughly linearly, we must assume that y_n is too. This gives us $y_n \approx n\varphi^2$. The approximations for x_n and y_n become exact as n approaches ∞ , and $y_n = x_n + n$, so we have $n\varphi^2 = n\varphi + n$, o.e. $\varphi^2 - \varphi - 1 = 0$. This gives $\varphi = \frac{1 \pm \sqrt{5}}{2}$, and this value must be positive so we take $\varphi = \frac{1+\sqrt{5}}{2}$.

When n does not approach infinity, the approximation $(x_n, y_n) \approx (n\varphi, n\varphi^2)$ does not give integers. To obtain integers, it is reasonable to guess that $(x_n, y_n) = (\lfloor n\varphi \rfloor, \lfloor n\varphi^2 \rfloor)$, as this preserves $y_n = \lfloor n\varphi^2 \rfloor = \lfloor n\varphi + n \rfloor = \lfloor n\varphi \rfloor + n = x_n + n$, x_n and y_n are strictly increasing, and the limiting ratios for x_n and y_n that we found above. Such sequences are called Beatty sequences, and have useful properties. Indeed, this solution gives the first few losing positions exactly, so we will now attempt to prove that it gives all of them.

All that we have left to show is that x_n indeed obtains $\text{mex}(x_0, y_0, \dots, x_{n-1}, y_{n-1})$, which given that x_n and y_n are strictly increasing, is equivalent to showing that $\lfloor n\varphi \rfloor$ and $\lfloor n\varphi^2 \rfloor$ partition the positive integers into 2 subsets. This is an application of Beatty's theorem, which we will prove for this case.

First we show that the images of the sequences are disjoint, i.e. $x_n \neq y_m$ for all $n, m \in \mathbb{Z}^+$. We can prove this by contradiction: first assume that there exist n, m such that $\lfloor n\varphi \rfloor = \lfloor m\varphi^2 \rfloor$. Then, by definition of the floor function, there exists $k \in \mathbb{Z}^+$ such that $k < n\varphi < k+1$ and $k < m\varphi^2 < k+1$. Note that the inequalities are strict because n, m are positive integers and φ, φ^2 are irrational. Summing the two inequalities, we obtain $\frac{k}{\varphi} + \frac{k}{\varphi^2} < n + m < \frac{k}{\varphi} + \frac{k}{\varphi^2} + 1$. By the

characteristic equation $\varphi^2 - \varphi - 1 = 0$, o.e. $\frac{1}{\varphi} + \frac{1}{\varphi^2} = 1$, $k < n + m < k + 1$. This is a contradiction: $n + m$ is an integer and as such cannot sit between two consecutive integers, so the images of the sequences are indeed disjoint.

Now all we have left to show is that all integers are in one of the subsets. We prove this by contradiction: assume there exists a positive integer k such that both x_n and y_m skip over k , i.e. there exist n, m, k such that $\lfloor n\varphi \rfloor < k$, $k + 1 \leq \lfloor (n + 1)\varphi \rfloor$, $\lfloor m\varphi^2 \rfloor < k$ and $k + 1 \leq \lfloor (m + 1)\varphi^2 \rfloor$. By the same reasons as in the above paragraph, we obtain $n\varphi < k$, $k + 1 < (n + 1)\varphi$, $m\varphi^2 < k$ and $k + 1 < (m + 1)\varphi^2$. This gives us $n < k/\varphi$, $(k + 1)/\varphi < n + 1$, $m < k/\varphi^2$ and $(k + 1)/\varphi^2 < m + 1$. Summing the corresponding pairs yields $n + m < k$ and $k + 1 < n + m + 2$, which resolves to the singular inequality $n + m < k < n + m + 1$. As before we have an integer between consecutive integers, so this is a contradiction. $\square$